Algebra Review: Unit 1 — Expressions & Equations

Station 1 of 4: Evaluating and Simplifying Expressions

Name: _____ Date: _____

Show your work. Simplify all answers completely.

1. Evaluate $3x^2 - 2x + 5$ for $x = -2$.

2. Evaluate $\dfrac{2m - n}{4}$ for $m = 9$ and $n = -2$.

3. Evaluate: $\dfrac{18}{3} \cdot 2 + 4^2 - 5$

4. Simplify: $4\,(2x - 3) + 5x$

5. Simplify: $7x - 3\,(x - 4) + 2$

6. Simplify: $-2(3y + 1) - (y - 5)$

7. Simplify: $\dfrac{1}{2}(8x - 6) + 3x$

8. Combine like terms: $5a + 3b - 2a + 7b - b$

9. Evaluate $\sqrt{x^2 + y^2}$ for $x = 5$ and $y = 12$.

10. Write an algebraic expression for *"5 less than three times a number n.*

Algebra Review: Unit 1 — Expressions & Equations

Station 2 of 4: One- and Two-Step Equations

Name: _____ Date: _____

Solve each equation. Check your solution.

1. Solve: $x + 9 = 4$

2. Solve: $y - 7 = -3$

3. Solve: $6n = -42$

4. Solve: $\frac{m}{5} = -3$

5. Solve: $3x + 5 = 20$

6. Solve: $4y - 9 = 15$

7. Solve: $\frac{m}{3} - 2 = 4$

8. Solve: $-2n + 7 = -11$

9. Solve: $\frac{x + 4}{5} = 3$

10. A gym charges a \$12 joining fee plus \$5 per visit. If Taylor has paid \$47 in total, how many visits v has Taylor made? Write and solve an equation.

Algebra Review: Unit 1 — Expressions & Equations

Station 3 of 4: Multi-Step Equations and Variables on Both Sides

Name: _____ Date: _____

Solve each equation. If there is no solution or all real numbers are solutions, say so.

1. Solve: $5x - 3 = 2x + 12$

2. Solve: $3\,(x + 2) = 21$

3. Solve: $4\,(2y - 1) = 6y + 8$

4. Solve: $7 - 2(x - 3) = x + 1$

5. Solve: $2x + 3x - 4 = 21$

6. Solve: $\frac{x}{2} + \frac{x}{3} = 10$

7. Solve: $\frac{2x - 1}{3} = \frac{x + 4}{2}$

8. Solve: $3\,(x - 4) = 3x + 5$

9. Solve: $2\,(3x + 1) = 6x + 2$

10. Solve: $0.5x + 1.2 = 0.2x - 0.3$

Algebra Review: Unit 1 — Expressions & Equations

Station 4 of 4: Literal Equations and Word Problems

Name: _____ Date: _____

Solve each literal equation for the indicated variable. Define a variable and write an equation for each word problem.

1. Solve for r: $C = 2\pi r$

2. Solve for h: $V = \frac{1}{3}\pi r^2 h$

3. Solve for x: $y = mx + b$

4. Solve for w: $P = 2\ell + 2w$

5. Solve for b: $A = \frac{1}{2}(a + b)h$

6. Solve for y: $3x + 4y = 12$

7. Three times a number, decreased by 8, equals 12 more than the number. Find the number.

8. The sum of three consecutive integers is 87. Find the integers.

9. The length of a rectangle is 3 more than twice its width. The perimeter is 36 cm. Find the length and width.

10. Plan A costs \$20 per month plus \$0.10 per minute. Plan B costs \$5 per month plus \$0.25 per minute. For how many minutes m do the plans cost the same?

Algebra Review: Unit 1 — Expressions & Equations

Answer Key

Station 1: Evaluating and Simplifying Expressions

1. 21 $\quad 3(4)+4+5$
2. 5 $\quad \dfrac{18+2}{4}$
3. 23 $\quad 12+16-5$
4. $13x-12$
5. $4x+14$
6. $-7y+3$
7. $7x-3$
8. $3a+9b$
9. 13 $\quad \sqrt{25+144}=\sqrt{169}$
10. $3n-5$

Station 2: One- and Two-Step Equations

1. $x=-5$
2. $y=4$
3. $n=-7$
4. $m=-15$
5. $x=5$
6. $y=6$
7. $m=18$
8. $n=9$
9. $x=11$
10. $12+5v=47;\ v=7$ visits

Station 3: Multi-Step Equations and Variables on Both Sides

1. $x=5$
2. $x=5$
3. $y=6$
4. $x=4$
5. $x=5$
6. $x=12$
7. $x=14$
8. no solution
9. all real numbers
10. $x=-5$

Station 4: Literal Equations and Word Problems

1. $r=\dfrac{C}{2\pi}$
2. $h=\dfrac{3V}{\pi r^2}$
3. $x=\dfrac{y-b}{m}$
4. $w=\dfrac{P-2\ell}{2}$
5. $b=\dfrac{2A}{h}-a$
6. $y=\dfrac{12-3x}{4}$
7. $3n-8=n+12;\ n=10$
8. 28, 29, 30 $\quad n+(n+1)+(n+2)=87$
9. width $=5$ cm, length $=13$ cm $\quad 2(2w+3)+2w=36$
10. $20+0.10m=5+0.25m;\ m=100$ minutes